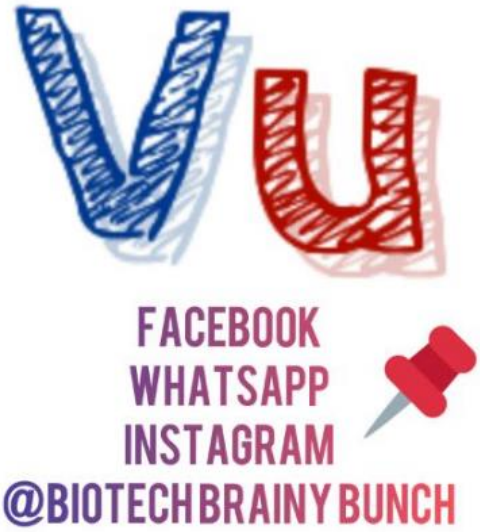


BIO201 Solved Final Papers Fall 2020

Biotech Brainy Bunch

Email; mehwishshabirhussain07@gmail.com

Follow on FaceBook and Instagram

@Biotechbrainybunch

EMAIL; to get added in Whatsapp Group

EFFORT BY; MEHWISH MUGHAL

Types of growth promoting oncogenes (5marks)

Autonomous Cell Growth in Cancer Cells Is Promoted By Following Elements

a) Growth Factors b) Growth Factor Receptors c) Signal-Transducing Proteins (secondary messengers) d) Nuclear Transcription Factors e) Cyclins and Cyclin Dependent Kinases (CDKs)

a) Growth Factors Some cancer cells acquire the ability through mutations to produce growth factors to which they are responsive (autocrine signaling). Many glioblastomas secrete PDGF and sarcomas make TGF- α. Similar loops are fairly common many types of cancers.

b) Growth Factor Receptors Mutated Growth Factor Receptors continue to deliver continues signals in the absence of the ligand. EGF receptor ERBB1 is over expressed in 80% of SCC of lung. ERBB2/HER2 is amplified in 25%-30% of breast cancers and adenocarcinomas of lung , ovary and salivary glands. These tumours are sensitive to small amounts of growth factors.

c) Signal-Transducing Proteins Mutaion of RAS is the most common oncogene abnormality, approximately 30% of all human tumors contain mutated version of RAS gene. GAP/neurofibramin 1 (a RAS deactivating protein) is mutated in neurofibramatosis type 1. BCR-ABL hybrid gene activates RAS pathways (like p53 Normal ABL gene also promotes apoptosis via NLS domain).

d) Nuclear Transcription Factors MYC and the polymerase assemble on a specific DNA sequence and drive transcription of genes. MYC protooncogene is expressed in virtually all the cells, and the MYC protein is rapidly induced when cells receive a signal to divide. MYC binds DNA and causes transcriptional activation of many growth related proteins. Dysregulation of of MYC gene occurs in Burkitt Lymphoma (B-cell tumor); MYC is also amplified in many other cancers. One effect of this pathway is the increased production of the gene regulatory protein Myc. Myc increases the transcription of several genes, including the gene encoding cyclin D.

e) Growth Promoting Oncogenes Cyclins & CDKs CDKs Phosphorilate crucial target proteins & are expressed constitutively during cell cycle in inactive form however cyclins are synthesized during specific phases of cell cycle.

LDL particles (3 marks)

Receptor-mediated endocytosis is the method by which cholesterol is taken up by most mammalian cells. Water-insoluble cholesterol and triglycerides are packaged by liver cells into lipoprotein particles, which are then secreted into the bloodstream to provide body tissues with lipids. One type of these particles, called low-density lipoproteins, or LDLs, must be taken up by the liver for recycling. This uptake also occurs via receptor-mediated endocytosis. This process begins with the binding of LDLs to specific receptor proteins on the cell surface. Once engulfed by endocytosis, the LDL particle is freed from the receptors. The receptors segregate to a region of the vesicle that buds off to form a new vesicle, which is recycled to the plasma membrane. The freed LDL particle remains in the original vesicle, which fuses with a lysosome in which the LDL is digested and the cholesterol made available for cell use. Persons with the inherited disease familial hypercholesterolemia (-emia, "blood") have dangerously high levels of cholesterol in their blood because of a deficient receptor for LDL.

cloning (2 marks)

Cloning is the process of generating a genetically identical copy of a cell or an organism. **Cloning** happens all the time in nature. In biomedical research, **cloning** is broadly defined to mean the duplication of any kind of biological material for scientific study, such as a piece of DNA or an individual cell.

Gene regulatory expression or housekeeping cells (3 marks)

There are two alternatives—transcriptional regulation and posttranscriptional regulation—can be distinguished by examining the actual RNA sequences made within the nucleus of each cell type. Such analyses indicate that for some proteins, the mechanism of regulation is differential gene transcription. Both brain and liver cells, for example, transcribe "housekeeping" genes—those that encode proteins involved in the basic metabolic processes that occur in every living cell, such as glycolysis enzymes. But liver cells transcribe some genes for liver-specific proteins, and brain cells transcribe some genes for brain-specific proteins. And neither cell type transcribes the genes for proteins that are characteristic of muscle, blood, bone, or the other specialized cell types in the body.

What is meant by polarity of water molecules? (3 marks)

water molecules interact with one another through polarity-induced hydrogen bonds, any molecule that is itself polar will interact with other polar molecules by weak (δ+ to δ–) attractions in hydrogen bonds. If a polar molecule interacts with water in this way, it is called hydrophilic ("water-loving").

Name of antibodies (2 marks)

- **IgG**, **IgM**, **IgA**, **IgE**, **IgD** and Camelid antibodies.

Function of T-cells (3 marks)

helper T cells are crucial for defense against both extracellular and intracellular pathogens. They help stimulate B cells to make antibodies that help inactivate or eliminate extracellular pathogens and their toxic products. They activate macrophages to destroy any intracellular pathogen multiplying within the macrophage's phagosomes, and they help activate cytotoxic T cells to kill infected target cells. Once a helper T cell has been activated by an antigen-presenting cell to become an effector cell, it can then help activate other cells. It does this both by secreting a variety of cytokines and by displaying costimulatory proteins on its surface.

Define epigenetic (2 marks)

the study of changes in organisms caused by modification of gene expression rather than alteration of the genetic code itself. Epigenetic information in the form of at least one distinct protein that preexists in the organelle membrane is also required, and this information is passed from parent cell to progeny cell in the form of the organelle itself. Presumably, such information is essential for the propagation of the cell's compartmental organization, just as the information in DNA is essential for the propagation of the cell's nucleotide and amino acid sequences.

TH1 and TH2 and their role (2 marks)

When activated by an antigen presenting cell, a naïve helper T cell can differentiate into either of two distinct types of effector helper cell, called TH1 and TH2. TH1 cells mainly help activate macrophages and cytotoxic T cells, whereas TH2 cells mainly help activate B cells.

Role of lipidbilayer (2 marks)

Phospholipid bilayers are critical components of **cell** membranes. The lipid bilayer acts as a barrier to the passage of molecules and ions into and out of the **cell**. However, an important function of the **cell membrane** is to allow selective passage of certain substances into and out of cells.

Signaling sequence (2 marks)

Signal sequences are N-terminal extensions of newly synthesized secretory and membrane proteins. ... In eukaryotes, **signal sequences** direct the insertion of proteins into the membrane of the endoplasmic reticulum and are usually cleaved off by **signal** peptidase.

What's stem cell? (2 marks) A stem cell is an undifferentiated cell that divides and differentiates into specialized cell(s).

Phagocytosis(2 marks): In phagocytosis ("cellular eating"), part of the plasma membrane engulfs large particles or even entire cells. Phagocytosis is used as a cellular feeding process by unicellular protists and by some white blood cells that defend the body by engulfing foreign cells and substances. The food vacuole or phagosome that forms usually fuses with a lysosome, where its contents are digested

Cytokinesis (2 marks): Cell separation, or cytokinesis, begins 20 minutes after chromosome replication is finished. The first event of cytokinesis is a pinching in of the plasma membrane to form a ring similar

to a purse string. Fibers composed of a protein similar to eukaryotic tubulin (which makes up microtubules) are major components of this ring. As the membrane pinches in, new cell wall materials are synthesized, which finally separate the two cells.

What is the role of mitochondria? (3 marks)

In eukaryotic cells, the breakdown of fuel molecules such as glucose begins in the cytosol. The molecules that result from this partial degradation enter the mitochondria (singular, mitochondrion), whose primary function is to convert the potential chemical energy of those fuel molecules into a form that the cell can use: the energy-rich molecule ATP (adenosine triphosphate). The production of ATP in the mitochondria using fuel molecules and molecular oxygen (O2) is called cellular respiration.

What is cytoskeleton? (3 marks)

Cytoplasm contains a set of long, thin fibers called the cytoskeleton. The cytoskeleton fills at least three important roles:

- It maintains cell shape and support.
- It provides for various types of cellular movement.
- Some of its fibers act as tracks or supports for motor proteins, which help move things within the cell.

There are three components of the cytoskeleton: microfilaments, intermediate filaments, and microtubules

What is operon? (2 marks)

The whole unit, consisting of the closely linked structural genes and the DNA sequences that control their transcription, is called an operon. An operon always consists of a promoter, an operator, and two or more structural genes. The promoter and operator are binding sites on DNA and are not transcribed.

RNA processing (2 marks)

Eukaryotic protein-coding genes contain some sequences that do not appear in the mature mRNA that is translated into proteins. To produce the mature mRNA, the primary transcript (pre-mRNA) is processed in several ways: introns are removed, exons are joined, and bases are added at both ends. This process is called RNA processing.

Differentiate between Euchromatin and Heterochromatin (3 marks)

Under a microscope, two kinds of chromatin can be distinguished in the stained interphase nucleus: Euchromatin and heterochromatin. Euchromatin is diffuse and stains lightly; it contains the DNA that is transcribed into mRNA. Heterochromatin stains densely and is generally not transcribed; any genes that it contains are thus inactivated.

Why we need cell culture in Vitro? (2 marks)

Essentially, **cell culture** involves the distribution of **cells** in an artificial environment (in **vitro**) which **is** composed of the necessary nutrients, ideal temperature, gases, pH and humidity to allow the **cells** to **grow** and proliferate

Difference between Homotypic and Heterotypic. (3marks)

In most cases, the binding of cells in a tissue is Homotypic; that is, the same molecule sticks out of both cells, and the exposed surfaces bind to each other. But heterotypic binding (of cells with different proteins) also can occur. For example, when the mammalian sperm meets the egg, different proteins on the two types of cells have complementary binding surfaces.

What was the technique used by Mendel in his experiment (2)

Emasculation and **bagging** were the techniques used by Mendel to perform hybridisation. If the female flower is bisexual, removal of anthers from the flower bud before the **anther**, dehisce using a pair of forceps. This is called as **emasculation**.

Steps of mitosis (3)

Mitosis is used to segregate them into the two new nuclei. The reproduction of a eukaryotic cell typically consists of three steps:

- The replication of DNA within the nucleus
- The packaging and segregation of the replicated DNA into two new nuclei (nuclear division)
- The division of the cytoplasm (cytokinosis)

How DNA is packaged in Cell. (2 marks)

DNA is tightly packed up to fit in the nucleus of every **cell**. As shown in the animation, a **DNA** molecule wraps around histone proteins to form tight loops called nucleosomes. These nucleosomes coil and stack together to form fibers called chromatin.

What is primary culture? (2 marks)

Primary culture refers to the stage of the **culture** after the cells are isolated from the tissue and proliferated under the appropriate conditions until they occupy all of the available substrate (i.e., reach confluence). ... After the first subculture, the **primary culture** becomes known as a cell line or subclone.

Why the cell is important? (3 marks)

Cells are the basic building blocks of all living things. The human body is composed of trillions of **cells**. They provide structure for the body, take in nutrients from food, convert those nutrients into energy, and carry out specialized functions.

Name three RNA viruses (3 marks)

Human diseases causing RNA viruses include:

Orthomyxoviruses, Hepatitis C Virus (HCV), Ebola disease, SARS, influenza, polio measles and retrovirus including adult Human T-cell lymphotropic virus type 1 (HTLV-1) and human immunodeficiency virus (HIV).

Which antibody is present on the surface of b cells? (2 marks)

IgM is not only the first class of antibody to appear on the surface of a developing B cell. It is also the major class secreted into the blood in the early stages of a primary antibody response, on first exposure to an **antigen**.

Missense mutation (2 marks)

In genetics, a **missense mutation** is a point **mutation** in which a single nucleotide change results in a codon that codes for a different amino acid. It is a type of nonsynonymous substitution.

what is palindromic sequence (3 marks)

A palindromic sequence is a nucleic acid sequence in a double-stranded DNA or RNA molecule wherein reading in a certain direction on one strand matches the sequence reading in the opposite direction on the complementary strand.

Active and passive transport (3 marks)

Active transport is the movement of molecules or ions against a concentration gradient (from an area of lower to higher concentration), which does not ordinarily occur, so enzymes and energy are required. **Passive transport** is the movement of molecules or ions from an area of higher to lower concentration.

Simple staining and gram staining (2 marks)

In **simple staining**, the bacterial smear is **stained** with a single reagent, which produces a distinctive contrast between the organism and its background. **Basic stains** with a positively charged chromogen are preferred because bacterial.

Gram stain or Gram staining, also called Gram's method, is a method of staining used to distinguish and classify bacterial species into two large groups: gram-positive bacteria and gram-negative bacteria.

What Is CANNA PK 13/14? (3 marks)

CANNA **PK 13**/14 is a flowering stimulator high in phosphorus and potassium manufactured by CANNA. **PK 13**/14 ensures stronger cell formation, the production of enough sugars in the flowering stage, and increases plants osmotic value which improves the volume, weight and taste of your crop.

Difference between Nucleotide and Nucleoside

Nucleotide	Nucleoside
The chemical composition of nucleotide consists of a phosphate group, a sugar and a nitrogenous base.	A nucleoside has a chemical composition that consists of a sugar and a base without the phosphate group.
They are one of the major causes of cancer-causing agents to this very day.	They are used as agents in medicine that are primarily used against viruses and cancer-causing agents.
Some of the major examples of nucleotides are adenosine, guanosine etc.	Some of the key examples of nucleosides are the same as nucleotides only with the addition of phosphate groups.

MHC and its types

There are two major **types** of **MHC** protein molecules—class I and class II. Class I **MHC** molecules span the membrane of almost every cell in an organism, while class II molecules are restricted to cells of the immune system called macrophages and lymphocytes.

Difference b/w MHC class I and MHC class II

MHC Class I	MHC Class II
Distribution; Present on almost all cells which have a nucleus	Has a comparatively limited distribution. Typically found in dendritic cells, macrophages, and B-cells
Encoding genes; HLA-A, HLA-B and HLA-C	HLA-D
Function; Presence of abundant antigens targets the cell for destruction	Presence of foreign antigens induce the production of antibodies
Recognized by CD8+ receptor of cytotoxic T cell	Recognized by CD4+ receptor of helper T cell
Enzymes involved; Cytosolic Proteasome	Lysosomal and endosomal proteases
Endogenous origins	Derived from extracellular proteins

Difference b/w purines and pyrimidines

Purines	Pyrimidines
Purine is a heterocyclic aromatic organic compound composed of a pyrimidine ring fused with imidazole ring.	Pyrimidine is a heterocyclic aromatic organic compound that is composed of carbon and hydrogen.
It comprises adenine and guanine as nucleobases.	It comprises Cytosine, thymine, uracil as nucleobases
It consists of two hydrogen-carbon rings and four nitrogen atoms	It consists of one hydrogen-carbon ring and two nitrogen atoms
The melting point of purine is 214 °C	The melting point of pyrimidine is 20-22 °C
Catabolism results in the production of uric acid	Catabolism produces carbon dioxide, beta-amino acids and ammonia

Difference b/w saturated and unsaturated fatty acids

Saturated fatty acids lack double bonds **between** the individual carbon atoms, while in **unsaturated fatty acids** there is at least one double bond **in the fatty acid** chain. **Saturated fats** tend to be solid at room temperature and from animal sources, while **unsaturated fats** are usually liquid and from plant sources.

Role of DNA in living organism

DNA is vital for all **living** beings – even plants. It is important for inheritance, coding for proteins and the genetic instruction guide for life and its processes. **DNA** holds the instructions for an **organism's** or each cell's development and reproduction and ultimately death.

Role of protein in living organism

Protein is a nutrient your body needs to grow, as well as to support and maintain your life. ... For example, **proteins** help cells develop and communicate, act as enzymes and hormones,

conduct the transport of nutrients throughout your bloodstream and repair damaged tissue. In other words, you can't live without **protein**.

How protein crosses double membrane structure of mitochondria.

Proteins are translocated into the **mitochondrial** matrix space by passing through the TOM and TIM complexes at sites of adhesion between the outer and inner **membranes** known as contact sites. ... A **second** set of hsp70 **proteins** in the matrix or stroma bind to the incoming polypeptide chain to pull it into the **organelle**.

Difference between negative and positive gram bacteria

Gram positive Bacteria	Gram Negative Bacteria
These bacteria retain the crystal violet colour even after they are washed with acetone or alcohol and appear as purple-coloured when examined under the microscope after gram staining	These bacteria do not retain the stain colour even after they are washed with acetone or alcohol and appear as pink-coloured when examined under the microscope after gram staining.

Process of developing of T-cell

T cells are derived from haematopoietic stem **cells** that are found in the bone marrow. The progenitors of these **cells** migrate **to** and colonise the thymus. ... The majority of **cells** in the thymus give rise **to** αβ **T cells**, however approximately 5% bear the γδ **T cell** receptor (TCR).

Cam kinase II function in cellular movement

CaM kinase II regulates neural excitability, phosphorylating Synapsin I (see Drosophila Synapsin), a protein which **functions** in regulating vesicular **movement**. Vesicles release neurotransmitters at the synapse, and their dynamics is the focus of regulation of neural excitability.

Function of Adaptive Immune system.

The adaptive immune system, also known as the specific immune system, is composed of highly-specialized systemic cells and processes that eliminate or prevent pathogenic **growth**. The adaptive immune system works to protect and heal the **body** when the innate immune system fails.

Rod cells (10 marks)

Rod, one of two types of photoreceptive cells in the retina of the eye in vertebrate animals. Rod cells function as specialized neurons that convert visual stimuli in the form of photons (particles of light) into chemical and electrical stimuli that can be processed by the central nervous system. Rod cells are stimulated by light over a wide range of intensities and are responsible for perceiving the size, shape, and brightness of visual images. They do not perceive colour and fine detail, tasks performed by the other major type of light-sensitive cell, the cone. Rod cells are

much more sensitive to light than cones and are also much more numerous. The human eye contains about 130 million rods and about 7 million cones. Rod cells have an elongated structure and consist of four distinct regions: an outer segment, an inner segment, the cell body, and the synaptic region. The outer segment contains the phototransduction apparatus. It is composed of a series of closely packed membrane disks that contain the photoreceptor molecule rhodopsin. Genetic mutations in the rhodopsin molecule have been shown to produce certain forms of retinitis pigmentosa, an inherited degenerative disease of the retinal pigments. The synaptic region is the site where the rod cell relays its information to intermediate neurons in the retina. These neurons connect with ganglion neurons whose axons form the approximately one million fibres of the optic nerve.

Rhodopsin is made up of a protein called opsin and a photosensitive chemical derived from vitamin A, 11-*cis*-retinaldehyde. Photons of light entering the eye cause the 11-*cis*-retinaldehyde to undergo isomerization (a change in configuration), forming all-*trans*-retinaldehyde. This isomerization activates the opsin protein, which then interacts with and activates a small protein called transducin. The association of opsin with transducin couples the external stimulus of light to an internal biochemical pathway that ultimately alters the release of neurotransmitters from the synaptic region of the cell. This changes the firing of the intermediate retinal neurons and affects the electrical impulses sent along the optic nerve to the brain. Such a complicated relay system allows for some integration and fine-tuning of these signals. Rhodopsin molecules are broken down in sunlight or other bright viewing conditions. This breakdown prevents the overstimulation of the rod cells and makes them less sensitive to a bright environment. In dim light there is little breakdown of rhodopsin, and its persisting high concentration allows for better vision in a dark environment. Dark-adapted vision in humans is basically devoid of colour because it depends almost entirely on the functioning of rods. *Compare* cone cells.

Clinical use of cell culture

Cell culture includes processes for isolating **cells** from **tissue** and maintaining, proliferating, and/or differentiating the **cells** in a **culture** vessel containing medium. ... Cultured **cells** are not only **used** for basic research in biology and molecular biology, but also for drug discovery and regenerative medicine.

Different stages of Cell cycle. (10 marks)

To divide, a cell must complete several important tasks: it must grow, copy its genetic material (DNA), and physically split into two daughter cells. Cells perform these tasks in an organized, predictable series of steps that make up the cell cycle. The cell cycle is a cycle, rather than a linear pathway, because at the end of each go-round, the two daughter cells can start the exact same process over again from the beginning.

In eukaryotic cells, or cells with a nucleus, the stages of the cell cycle are divided into two major phases: **interphase** and the **mitotic (M) phase**.

- During *interphase*, the cell grows and makes a copy of its DNA.
- During the *mitotic (M) phase*, the cell separates its DNA into two sets and divides its cytoplasm, forming two new cells.
- **S phase.** In S phase, the cell synthesizes a complete copy of the DNA in its nucleus. It also duplicates a microtubule-organizing structure called the centrosome. The centrosomes help separate DNA during M phase.

  M phase During the mitotic (M) phase, the cell divides its copied DNA and cytoplasm to make two new cells. M phase involves two distinct division-related processes: mitosis and cytokinesis.

  In **mitosis**, the nuclear DNA of the cell condenses into visible chromosomes and is pulled apart by the mitotic spindle, a specialized structure made out of microtubules. Mitosis takes place in four stages: prophase (sometimes divided into early prophase and prometaphase), metaphase, anaphase, and telophase. You can learn more about these stages in the video on mitosis.

  In **cytokinesis**, the cytoplasm of the cell is split in two, making two new cells. Cytokinesis usually begins just as mitosis is ending, with a little overlap. Importantly, cytokinesis takes place differently in animal and plant cells.